

NCERT CLASS 7

60 Questions Per Topic

Based on the refined 2026–27 curriculum map created for this project.

Each curriculum topic below contains exactly 60 questions: understanding, examples/classification, comparison, application, reasoning/error analysis, and higher-order activity.

These are question-bank items derived from the topic descriptions in the supplied curriculum map. Validate against the exact school/textbook edition before high-stakes use.

MATHEMATICS — 1. Large Numbers Around Us

Coverage: ■ Large-number representation; estimation; place value; real-world quantitative comparisons.

Understanding

1. What is ■ Large-number representation, and what are its main features? Use a simple example.
2. What is estimation, and what are its main features? Use a school-life example.
3. What is place value, and what are its main features? Use a real-world example.
4. What is real-world quantitative comparisons., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Large-number representation, and what are its main features? Explain in your own words.
6. What is estimation, and what are its main features? Give a step-by-step response.
7. What is place value, and what are its main features? Mention one common misconception.
8. What is real-world quantitative comparisons., and what are its main features? State one practical application.
9. What is ■ Large-number representation, and what are its main features? Justify your response.
10. What is estimation, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to place value and explain each. Use a simple example.
12. Give two examples related to real-world quantitative comparisons. and explain each. Use a school-life example.
13. Give two examples related to ■ Large-number representation and explain each. Use a real-world example.
14. Give two examples related to estimation and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to place value and explain each. Explain in your own words.
16. Give two examples related to real-world quantitative comparisons. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Large-number representation and explain each. Mention one common misconception.
18. Give two examples related to estimation and explain each. State one practical application.
19. Give two examples related to place value and explain each. Justify your response.
20. Give two examples related to real-world quantitative comparisons. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Large-number representation related to estimation? Compare the two ideas. Use a simple example.
22. How is estimation related to place value? Compare the two ideas. Use a school-life example.
23. How is place value related to real-world quantitative comparisons.? Compare the two ideas. Use a real-world example.
24. How is real-world quantitative comparisons. related to ■ Large-number representation? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Large-number representation related to estimation? Compare the two ideas. Explain in your own words.
26. How is estimation related to place value? Compare the two ideas. Give a step-by-step response.
27. How is place value related to real-world quantitative comparisons.? Compare the two ideas. Mention one common misconception.
28. How is real-world quantitative comparisons. related to ■ Large-number representation? Compare the two ideas. State one practical application.
29. How is ■ Large-number representation related to estimation? Compare the two ideas. Justify your response.
30. How is estimation related to place value? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply place value to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply real-world quantitative comparisons. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Large-number representation to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply estimation to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply place value to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply real-world quantitative comparisons. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Large-number representation to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply estimation to a realistic situation. What would you do or calculate? State one practical application.
39. Apply place value to a realistic situation. What would you do or calculate? Justify your response.
40. Apply real-world quantitative comparisons. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Large-number representation. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of estimation. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of place value. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of real-world quantitative comparisons.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Large-number representation. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of estimation. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of place value. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of real-world quantitative comparisons.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Large-number representation. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of estimation. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show place value. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show real-world quantitative comparisons.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Large-number representation. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show estimation. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show place value. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show real-world quantitative comparisons.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Large-number representation. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show estimation. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show place value. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show real-world quantitative comparisons.. Include the relevant rule or principle.

MATHEMATICS — 2. Arithmetic Expressions

Coverage: ■ Order of operations; brackets; simplifying numerical expressions; reasoning.

Understanding

1. What is ■ Order of operations, and what are its main features? Use a simple example.
2. What is brackets, and what are its main features? Use a school-life example.
3. What is simplifying numerical expressions, and what are its main features? Use a real-world example.
4. What is reasoning., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Order of operations, and what are its main features? Explain in your own words.
6. What is brackets, and what are its main features? Give a step-by-step response.
7. What is simplifying numerical expressions, and what are its main features? Mention one common misconception.
8. What is reasoning., and what are its main features? State one practical application.
9. What is ■ Order of operations, and what are its main features? Justify your response.
10. What is brackets, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to simplifying numerical expressions and explain each. Use a simple example.
12. Give two examples related to reasoning. and explain each. Use a school-life example.
13. Give two examples related to ■ Order of operations and explain each. Use a real-world example.
14. Give two examples related to brackets and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to simplifying numerical expressions and explain each. Explain in your own words.
16. Give two examples related to reasoning. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Order of operations and explain each. Mention one common misconception.
18. Give two examples related to brackets and explain each. State one practical application.
19. Give two examples related to simplifying numerical expressions and explain each. Justify your response.
20. Give two examples related to reasoning. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Order of operations related to brackets? Compare the two ideas. Use a simple example.
22. How is brackets related to simplifying numerical expressions? Compare the two ideas. Use a school-life example.
23. How is simplifying numerical expressions related to reasoning.? Compare the two ideas. Use a real-world example.
24. How is reasoning. related to ■ Order of operations? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Order of operations related to brackets? Compare the two ideas. Explain in your own words.
26. How is brackets related to simplifying numerical expressions? Compare the two ideas. Give a step-by-step response.
27. How is simplifying numerical expressions related to reasoning.? Compare the two ideas. Mention one common misconception.
28. How is reasoning. related to ■ Order of operations? Compare the two ideas. State one practical application.
29. How is ■ Order of operations related to brackets? Compare the two ideas. Justify your response.
30. How is brackets related to simplifying numerical expressions? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply simplifying numerical expressions to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply reasoning. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Order of operations to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply brackets to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply simplifying numerical expressions to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply reasoning. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Order of operations to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply brackets to a realistic situation. What would you do or calculate? State one practical application.
39. Apply simplifying numerical expressions to a realistic situation. What would you do or calculate? Justify your response.
40. Apply reasoning. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Order of operations. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of brackets. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of simplifying numerical expressions. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of reasoning.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Order of operations. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of brackets. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of simplifying numerical expressions. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of reasoning.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Order of operations. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of brackets. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show simplifying numerical expressions. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show reasoning.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Order of operations. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show brackets. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show simplifying numerical expressions. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show reasoning.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Order of operations. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show brackets. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show simplifying numerical expressions. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show reasoning.. Include the relevant rule or principle.

MATHEMATICS — 3. A Peek Beyond the Point

Coverage: ■ Decimals; place value; operations; measurement applications.

Understanding

1. What is ■ Decimals, and what are its main features? Use a simple example.
2. What is place value, and what are its main features? Use a school-life example.
3. What is operations, and what are its main features? Use a real-world example.
4. What is measurement applications., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Decimals, and what are its main features? Explain in your own words.
6. What is place value, and what are its main features? Give a step-by-step response.
7. What is operations, and what are its main features? Mention one common misconception.
8. What is measurement applications., and what are its main features? State one practical application.
9. What is ■ Decimals, and what are its main features? Justify your response.
10. What is place value, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to operations and explain each. Use a simple example.
12. Give two examples related to measurement applications. and explain each. Use a school-life example.
13. Give two examples related to ■ Decimals and explain each. Use a real-world example.
14. Give two examples related to place value and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to operations and explain each. Explain in your own words.
16. Give two examples related to measurement applications. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Decimals and explain each. Mention one common misconception.
18. Give two examples related to place value and explain each. State one practical application.
19. Give two examples related to operations and explain each. Justify your response.
20. Give two examples related to measurement applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Decimals related to place value? Compare the two ideas. Use a simple example.
22. How is place value related to operations? Compare the two ideas. Use a school-life example.
23. How is operations related to measurement applications.? Compare the two ideas. Use a real-world example.
24. How is measurement applications. related to ■ Decimals? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Decimals related to place value? Compare the two ideas. Explain in your own words.
26. How is place value related to operations? Compare the two ideas. Give a step-by-step response.
27. How is operations related to measurement applications.? Compare the two ideas. Mention one common misconception.
28. How is measurement applications. related to ■ Decimals? Compare the two ideas. State one practical application.
29. How is ■ Decimals related to place value? Compare the two ideas. Justify your response.
30. How is place value related to operations? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply operations to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply measurement applications. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Decimals to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply place value to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply operations to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply measurement applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Decimals to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply place value to a realistic situation. What would you do or calculate? State one practical application.
39. Apply operations to a realistic situation. What would you do or calculate? Justify your response.
40. Apply measurement applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Decimals. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of place value. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of operations. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of measurement applications.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Decimals. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of place value. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of operations. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of measurement applications.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Decimals. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of place value. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show operations. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show measurement applications.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Decimals. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show place value. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show operations. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show measurement applications.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Decimals. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show place value. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show operations. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show measurement applications.. Include the relevant rule or principle.

MATHEMATICS — 4. Expressions using Letter-Numbers

Coverage: ■ Variables; algebraic expressions; substitution; simple modelling.

Understanding

1. What is ■ Variables, and what are its main features? Use a simple example.
2. What is algebraic expressions, and what are its main features? Use a school-life example.
3. What is substitution, and what are its main features? Use a real-world example.
4. What is simple modelling., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Variables, and what are its main features? Explain in your own words.
6. What is algebraic expressions, and what are its main features? Give a step-by-step response.
7. What is substitution, and what are its main features? Mention one common misconception.
8. What is simple modelling., and what are its main features? State one practical application.
9. What is ■ Variables, and what are its main features? Justify your response.
10. What is algebraic expressions, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to substitution and explain each. Use a simple example.
12. Give two examples related to simple modelling. and explain each. Use a school-life example.
13. Give two examples related to ■ Variables and explain each. Use a real-world example.
14. Give two examples related to algebraic expressions and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to substitution and explain each. Explain in your own words.
16. Give two examples related to simple modelling. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Variables and explain each. Mention one common misconception.
18. Give two examples related to algebraic expressions and explain each. State one practical application.
19. Give two examples related to substitution and explain each. Justify your response.
20. Give two examples related to simple modelling. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Variables related to algebraic expressions? Compare the two ideas. Use a simple example.
22. How is algebraic expressions related to substitution? Compare the two ideas. Use a school-life example.
23. How is substitution related to simple modelling.? Compare the two ideas. Use a real-world example.
24. How is simple modelling. related to ■ Variables? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Variables related to algebraic expressions? Compare the two ideas. Explain in your own words.
26. How is algebraic expressions related to substitution? Compare the two ideas. Give a step-by-step response.
27. How is substitution related to simple modelling.? Compare the two ideas. Mention one common misconception.
28. How is simple modelling. related to ■ Variables? Compare the two ideas. State one practical application.
29. How is ■ Variables related to algebraic expressions? Compare the two ideas. Justify your response.
30. How is algebraic expressions related to substitution? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply substitution to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply simple modelling. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Variables to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply algebraic expressions to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply substitution to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply simple modelling. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Variables to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply algebraic expressions to a realistic situation. What would you do or calculate? State one practical application.
39. Apply substitution to a realistic situation. What would you do or calculate? Justify your response.
40. Apply simple modelling. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Variables. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of algebraic expressions. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of substitution. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of simple modelling.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Variables. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of algebraic expressions. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of substitution. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of simple modelling.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Variables. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of algebraic expressions. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show substitution. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show simple modelling.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Variables. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show algebraic expressions. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show substitution. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show simple modelling.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Variables. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show algebraic expressions. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show substitution. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show simple modelling.. Include the relevant rule or principle.

MATHEMATICS — 5. Parallel and Intersecting Lines

Coverage: ■ Parallel/perpendicular lines; transversals; corresponding/alternate/interior angle relationships.

Understanding

1. What is ■ Parallel/perpendicular lines, and what are its main features? Use a simple example.
2. What is transversals, and what are its main features? Use a school-life example.
3. What is corresponding/alternate/interior angle relationships., and what are its main features? Use a real-world example.
4. What is ■ Parallel/perpendicular lines, and what are its main features? Show the idea with a diagram where appropriate.
5. What is transversals, and what are its main features? Explain in your own words.
6. What is corresponding/alternate/interior angle relationships., and what are its main features? Give a step-by-step response.
7. What is ■ Parallel/perpendicular lines, and what are its main features? Mention one common misconception.
8. What is transversals, and what are its main features? State one practical application.
9. What is corresponding/alternate/interior angle relationships., and what are its main features? Justify your response.
10. What is ■ Parallel/perpendicular lines, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to transversals and explain each. Use a simple example.
12. Give two examples related to corresponding/alternate/interior angle relationships. and explain each. Use a school-life example.
13. Give two examples related to ■ Parallel/perpendicular lines and explain each. Use a real-world example.
14. Give two examples related to transversals and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to corresponding/alternate/interior angle relationships. and explain each. Explain in your own words.
16. Give two examples related to ■ Parallel/perpendicular lines and explain each. Give a step-by-step response.
17. Give two examples related to transversals and explain each. Mention one common misconception.
18. Give two examples related to corresponding/alternate/interior angle relationships. and explain each. State one practical application.
19. Give two examples related to ■ Parallel/perpendicular lines and explain each. Justify your response.
20. Give two examples related to transversals and explain each. Include the relevant rule or principle.

Comparison

21. How is corresponding/alternate/interior angle relationships. related to ■ Parallel/perpendicular lines? Compare the two ideas. Use a simple example.
22. How is ■ Parallel/perpendicular lines related to transversals? Compare the two ideas. Use a school-life example.
23. How is transversals related to corresponding/alternate/interior angle relationships.? Compare the two ideas. Use a real-world example.
24. How is corresponding/alternate/interior angle relationships. related to ■ Parallel/perpendicular lines? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Parallel/perpendicular lines related to transversals? Compare the two ideas. Explain in your own words.
26. How is transversals related to corresponding/alternate/interior angle relationships.? Compare the two ideas. Give a step-by-step response.
27. How is corresponding/alternate/interior angle relationships. related to ■ Parallel/perpendicular lines? Compare the two ideas. Mention one common misconception.
28. How is ■ Parallel/perpendicular lines related to transversals? Compare the two ideas. State one practical application.
29. How is transversals related to corresponding/alternate/interior angle relationships.? Compare the two ideas. Justify your response.
30. How is corresponding/alternate/interior angle relationships. related to ■ Parallel/perpendicular lines? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Parallel/perpendicular lines to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply transversals to a realistic situation. What would you do or calculate? Use a school-life example.

33. Apply corresponding/alternate/interior angle relationships. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Parallel/perpendicular lines to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply transversals to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply corresponding/alternate/interior angle relationships. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Parallel/perpendicular lines to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply transversals to a realistic situation. What would you do or calculate? State one practical application.
39. Apply corresponding/alternate/interior angle relationships. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Parallel/perpendicular lines to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of transversals. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of corresponding/alternate/interior angle relationships.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Parallel/perpendicular lines. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of transversals. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of corresponding/alternate/interior angle relationships.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Parallel/perpendicular lines. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of transversals. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of corresponding/alternate/interior angle relationships.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Parallel/perpendicular lines. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of transversals. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show corresponding/alternate/interior angle relationships.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Parallel/perpendicular lines. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show transversals. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show corresponding/alternate/interior angle relationships.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Parallel/perpendicular lines. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show transversals. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show corresponding/alternate/interior angle relationships.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Parallel/perpendicular lines. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show transversals. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show corresponding/alternate/interior angle relationships.. Include the relevant rule or principle.

MATHEMATICS — 6. Number Play

Coverage: ■ Divisibility patterns; number puzzles; logical deductions.

Understanding

1. What is ■ Divisibility patterns, and what are its main features? Use a simple example.
2. What is number puzzles, and what are its main features? Use a school-life example.
3. What is logical deductions., and what are its main features? Use a real-world example.
4. What is ■ Divisibility patterns, and what are its main features? Show the idea with a diagram where appropriate.
5. What is number puzzles, and what are its main features? Explain in your own words.
6. What is logical deductions., and what are its main features? Give a step-by-step response.
7. What is ■ Divisibility patterns, and what are its main features? Mention one common misconception.
8. What is number puzzles, and what are its main features? State one practical application.
9. What is logical deductions., and what are its main features? Justify your response.
10. What is ■ Divisibility patterns, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to number puzzles and explain each. Use a simple example.
12. Give two examples related to logical deductions. and explain each. Use a school-life example.
13. Give two examples related to ■ Divisibility patterns and explain each. Use a real-world example.
14. Give two examples related to number puzzles and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to logical deductions. and explain each. Explain in your own words.
16. Give two examples related to ■ Divisibility patterns and explain each. Give a step-by-step response.
17. Give two examples related to number puzzles and explain each. Mention one common misconception.
18. Give two examples related to logical deductions. and explain each. State one practical application.
19. Give two examples related to ■ Divisibility patterns and explain each. Justify your response.
20. Give two examples related to number puzzles and explain each. Include the relevant rule or principle.

Comparison

21. How is logical deductions. related to ■ Divisibility patterns? Compare the two ideas. Use a simple example.
22. How is ■ Divisibility patterns related to number puzzles? Compare the two ideas. Use a school-life example.
23. How is number puzzles related to logical deductions.? Compare the two ideas. Use a real-world example.
24. How is logical deductions. related to ■ Divisibility patterns? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Divisibility patterns related to number puzzles? Compare the two ideas. Explain in your own words.
26. How is number puzzles related to logical deductions.? Compare the two ideas. Give a step-by-step response.
27. How is logical deductions. related to ■ Divisibility patterns? Compare the two ideas. Mention one common misconception.
28. How is ■ Divisibility patterns related to number puzzles? Compare the two ideas. State one practical application.
29. How is number puzzles related to logical deductions.? Compare the two ideas. Justify your response.
30. How is logical deductions. related to ■ Divisibility patterns? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Divisibility patterns to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply number puzzles to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply logical deductions. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Divisibility patterns to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply number puzzles to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply logical deductions. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Divisibility patterns to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply number puzzles to a realistic situation. What would you do or calculate? State one practical application.
39. Apply logical deductions. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Divisibility patterns to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of number puzzles. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of logical deductions.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Divisibility patterns. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of number puzzles. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of logical deductions.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Divisibility patterns. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of number puzzles. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of logical deductions.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Divisibility patterns. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of number puzzles. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show logical deductions.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Divisibility patterns. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show number puzzles. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show logical deductions.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Divisibility patterns. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show number puzzles. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show logical deductions.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Divisibility patterns. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show number puzzles. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show logical deductions.. Include the relevant rule or principle.

MATHEMATICS — 7. A Tale of Three Intersecting Lines

Coverage: ■ Triangles; classification; angle-sum property; geometric reasoning.

Understanding

1. What is ■ Triangles, and what are its main features? Use a simple example.
2. What is classification, and what are its main features? Use a school-life example.
3. What is angle-sum property, and what are its main features? Use a real-world example.
4. What is geometric reasoning., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Triangles, and what are its main features? Explain in your own words.
6. What is classification, and what are its main features? Give a step-by-step response.
7. What is angle-sum property, and what are its main features? Mention one common misconception.
8. What is geometric reasoning., and what are its main features? State one practical application.
9. What is ■ Triangles, and what are its main features? Justify your response.
10. What is classification, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to angle-sum property and explain each. Use a simple example.
12. Give two examples related to geometric reasoning. and explain each. Use a school-life example.
13. Give two examples related to ■ Triangles and explain each. Use a real-world example.
14. Give two examples related to classification and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to angle-sum property and explain each. Explain in your own words.
16. Give two examples related to geometric reasoning. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Triangles and explain each. Mention one common misconception.
18. Give two examples related to classification and explain each. State one practical application.
19. Give two examples related to angle-sum property and explain each. Justify your response.
20. Give two examples related to geometric reasoning. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Triangles related to classification? Compare the two ideas. Use a simple example.
22. How is classification related to angle-sum property? Compare the two ideas. Use a school-life example.
23. How is angle-sum property related to geometric reasoning.? Compare the two ideas. Use a real-world example.
24. How is geometric reasoning. related to ■ Triangles? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Triangles related to classification? Compare the two ideas. Explain in your own words.
26. How is classification related to angle-sum property? Compare the two ideas. Give a step-by-step response.
27. How is angle-sum property related to geometric reasoning.? Compare the two ideas. Mention one common misconception.
28. How is geometric reasoning. related to ■ Triangles? Compare the two ideas. State one practical application.
29. How is ■ Triangles related to classification? Compare the two ideas. Justify your response.
30. How is classification related to angle-sum property? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply angle-sum property to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply geometric reasoning. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Triangles to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply classification to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply angle-sum property to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply geometric reasoning. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Triangles to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply classification to a realistic situation. What would you do or calculate? State one practical application.
39. Apply angle-sum property to a realistic situation. What would you do or calculate? Justify your response.
40. Apply geometric reasoning. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Triangles. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of classification. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of angle-sum property. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of geometric reasoning.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Triangles. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of classification. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of angle-sum property. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of geometric reasoning.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Triangles. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of classification. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show angle-sum property. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show geometric reasoning.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Triangles. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show classification. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show angle-sum property. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show geometric reasoning.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Triangles. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show classification. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show angle-sum property. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show geometric reasoning.. Include the relevant rule or principle.

MATHEMATICS — 8. Working with Fractions

Coverage: ■ Fractions/mixed numbers; operations; comparison; contextual problems.

Understanding

1. What is ■ Fractions/mixed numbers, and what are its main features? Use a simple example.
2. What is operations, and what are its main features? Use a school-life example.
3. What is comparison, and what are its main features? Use a real-world example.
4. What is contextual problems., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Fractions/mixed numbers, and what are its main features? Explain in your own words.
6. What is operations, and what are its main features? Give a step-by-step response.
7. What is comparison, and what are its main features? Mention one common misconception.
8. What is contextual problems., and what are its main features? State one practical application.
9. What is ■ Fractions/mixed numbers, and what are its main features? Justify your response.
10. What is operations, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to comparison and explain each. Use a simple example.
12. Give two examples related to contextual problems. and explain each. Use a school-life example.
13. Give two examples related to ■ Fractions/mixed numbers and explain each. Use a real-world example.
14. Give two examples related to operations and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to comparison and explain each. Explain in your own words.
16. Give two examples related to contextual problems. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Fractions/mixed numbers and explain each. Mention one common misconception.
18. Give two examples related to operations and explain each. State one practical application.
19. Give two examples related to comparison and explain each. Justify your response.
20. Give two examples related to contextual problems. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Fractions/mixed numbers related to operations? Compare the two ideas. Use a simple example.
22. How is operations related to comparison? Compare the two ideas. Use a school-life example.
23. How is comparison related to contextual problems.? Compare the two ideas. Use a real-world example.
24. How is contextual problems. related to ■ Fractions/mixed numbers? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Fractions/mixed numbers related to operations? Compare the two ideas. Explain in your own words.
26. How is operations related to comparison? Compare the two ideas. Give a step-by-step response.
27. How is comparison related to contextual problems.? Compare the two ideas. Mention one common misconception.
28. How is contextual problems. related to ■ Fractions/mixed numbers? Compare the two ideas. State one practical application.
29. How is ■ Fractions/mixed numbers related to operations? Compare the two ideas. Justify your response.
30. How is operations related to comparison? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply comparison to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply contextual problems. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Fractions/mixed numbers to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply operations to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply comparison to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply contextual problems. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Fractions/mixed numbers to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply operations to a realistic situation. What would you do or calculate? State one practical application.
39. Apply comparison to a realistic situation. What would you do or calculate? Justify your response.
40. Apply contextual problems. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Fractions/mixed numbers. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of operations. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of comparison. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of contextual problems.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Fractions/mixed numbers. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of operations. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of comparison. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of contextual problems.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Fractions/mixed numbers. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of operations. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show comparison. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show contextual problems.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Fractions/mixed numbers. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show operations. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show comparison. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show contextual problems.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Fractions/mixed numbers. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show operations. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show comparison. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show contextual problems.. Include the relevant rule or principle.

MATHEMATICS — 9. Perimeter and Area

Coverage: ■ Area/perimeter of triangles and parallelograms; composite applications.

Understanding

1. What is ■ Area/perimeter of triangles and parallelograms, and what are its main features? Use a simple example.
2. What is composite applications., and what are its main features? Use a school-life example.
3. What is ■ Area/perimeter of triangles and parallelograms, and what are its main features? Use a real-world example.
4. What is composite applications., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Area/perimeter of triangles and parallelograms, and what are its main features? Explain in your own words.
6. What is composite applications., and what are its main features? Give a step-by-step response.
7. What is ■ Area/perimeter of triangles and parallelograms, and what are its main features? Mention one common misconception.
8. What is composite applications., and what are its main features? State one practical application.
9. What is ■ Area/perimeter of triangles and parallelograms, and what are its main features? Justify your response.
10. What is composite applications., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Area/perimeter of triangles and parallelograms and explain each. Use a simple example.
12. Give two examples related to composite applications. and explain each. Use a school-life example.
13. Give two examples related to ■ Area/perimeter of triangles and parallelograms and explain each. Use a real-world example.
14. Give two examples related to composite applications. and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to ■ Area/perimeter of triangles and parallelograms and explain each. Explain in your own words.
16. Give two examples related to composite applications. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Area/perimeter of triangles and parallelograms and explain each. Mention one common misconception.
18. Give two examples related to composite applications. and explain each. State one practical application.
19. Give two examples related to ■ Area/perimeter of triangles and parallelograms and explain each. Justify your response.
20. Give two examples related to composite applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Area/perimeter of triangles and parallelograms related to composite applications.? Compare the two ideas. Use a simple example.
22. How is composite applications. related to ■ Area/perimeter of triangles and parallelograms? Compare the two ideas. Use a school-life example.
23. How is ■ Area/perimeter of triangles and parallelograms related to composite applications.? Compare the two ideas. Use a real-world example.
24. How is composite applications. related to ■ Area/perimeter of triangles and parallelograms? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Area/perimeter of triangles and parallelograms related to composite applications.? Compare the two ideas. Explain in your own words.
26. How is composite applications. related to ■ Area/perimeter of triangles and parallelograms? Compare the two ideas. Give a step-by-step response.
27. How is ■ Area/perimeter of triangles and parallelograms related to composite applications.? Compare the two ideas. Mention one common misconception.
28. How is composite applications. related to ■ Area/perimeter of triangles and parallelograms? Compare the two ideas. State one practical application.
29. How is ■ Area/perimeter of triangles and parallelograms related to composite applications.? Compare the two ideas. Justify your response.
30. How is composite applications. related to ■ Area/perimeter of triangles and parallelograms? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Area/perimeter of triangles and parallelograms to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply composite applications. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Area/perimeter of triangles and parallelograms to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply composite applications. to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply ■ Area/perimeter of triangles and parallelograms to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply composite applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Area/perimeter of triangles and parallelograms to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply composite applications. to a realistic situation. What would you do or calculate? State one practical application.
39. Apply ■ Area/perimeter of triangles and parallelograms to a realistic situation. What would you do or calculate? Justify your response.
40. Apply composite applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Area/perimeter of triangles and parallelograms. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of composite applications.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Area/perimeter of triangles and parallelograms. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of composite applications.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Area/perimeter of triangles and parallelograms. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of composite applications.. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of ■ Area/perimeter of triangles and parallelograms. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of composite applications.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Area/perimeter of triangles and parallelograms. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of composite applications.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area/perimeter of triangles and parallelograms. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show composite applications.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area/perimeter of triangles and parallelograms. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show composite applications.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area/perimeter of triangles and parallelograms. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show composite applications.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area/perimeter of triangles and parallelograms. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show composite applications.. State one practical application.

- 59.** Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area/perimeter of triangles and parallelograms. Justify your response.
- 60.** Design a short activity, investigation, problem, model, or demonstration to test or show composite applications.. Include the relevant rule or principle.

MATHEMATICS — 10. Symmetry

Coverage: ■ Line/rotational symmetry; symmetry in figures and designs.

Understanding

1. What is ■ Line/rotational symmetry, and what are its main features? Use a simple example.
2. What is symmetry in figures and designs., and what are its main features? Use a school-life example.
3. What is ■ Line/rotational symmetry, and what are its main features? Use a real-world example.
4. What is symmetry in figures and designs., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Line/rotational symmetry, and what are its main features? Explain in your own words.
6. What is symmetry in figures and designs., and what are its main features? Give a step-by-step response.
7. What is ■ Line/rotational symmetry, and what are its main features? Mention one common misconception.
8. What is symmetry in figures and designs., and what are its main features? State one practical application.
9. What is ■ Line/rotational symmetry, and what are its main features? Justify your response.
10. What is symmetry in figures and designs., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Line/rotational symmetry and explain each. Use a simple example.
12. Give two examples related to symmetry in figures and designs. and explain each. Use a school-life example.
13. Give two examples related to ■ Line/rotational symmetry and explain each. Use a real-world example.
14. Give two examples related to symmetry in figures and designs. and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to ■ Line/rotational symmetry and explain each. Explain in your own words.
16. Give two examples related to symmetry in figures and designs. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Line/rotational symmetry and explain each. Mention one common misconception.
18. Give two examples related to symmetry in figures and designs. and explain each. State one practical application.
19. Give two examples related to ■ Line/rotational symmetry and explain each. Justify your response.
20. Give two examples related to symmetry in figures and designs. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Line/rotational symmetry related to symmetry in figures and designs.? Compare the two ideas. Use a simple example.
22. How is symmetry in figures and designs. related to ■ Line/rotational symmetry? Compare the two ideas. Use a school-life example.
23. How is ■ Line/rotational symmetry related to symmetry in figures and designs.? Compare the two ideas. Use a real-world example.
24. How is symmetry in figures and designs. related to ■ Line/rotational symmetry? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Line/rotational symmetry related to symmetry in figures and designs.? Compare the two ideas. Explain in your own words.
26. How is symmetry in figures and designs. related to ■ Line/rotational symmetry? Compare the two ideas. Give a step-by-step response.
27. How is ■ Line/rotational symmetry related to symmetry in figures and designs.? Compare the two ideas. Mention one common misconception.
28. How is symmetry in figures and designs. related to ■ Line/rotational symmetry? Compare the two ideas. State one practical application.
29. How is ■ Line/rotational symmetry related to symmetry in figures and designs.? Compare the two ideas. Justify your response.
30. How is symmetry in figures and designs. related to ■ Line/rotational symmetry? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Line/rotational symmetry to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply symmetry in figures and designs. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Line/rotational symmetry to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply symmetry in figures and designs. to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply ■ Line/rotational symmetry to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply symmetry in figures and designs. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Line/rotational symmetry to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply symmetry in figures and designs. to a realistic situation. What would you do or calculate? State one practical application.
39. Apply ■ Line/rotational symmetry to a realistic situation. What would you do or calculate? Justify your response.
40. Apply symmetry in figures and designs. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Line/rotational symmetry. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of symmetry in figures and designs.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Line/rotational symmetry. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of symmetry in figures and designs.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Line/rotational symmetry. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of symmetry in figures and designs.. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of ■ Line/rotational symmetry. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of symmetry in figures and designs.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Line/rotational symmetry. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of symmetry in figures and designs.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Line/rotational symmetry. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry in figures and designs.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Line/rotational symmetry. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry in figures and designs.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Line/rotational symmetry. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry in figures and designs.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Line/rotational symmetry. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry in figures and designs.. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Line/rotational symmetry. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry in figures and designs.. Include the relevant rule or principle.

MATHEMATICS — 11. The Circle

Coverage: ■ Radius; diameter; chords; circumference intuition; constructions.

Understanding

1. What is ■ Radius, and what are its main features? Use a simple example.
2. What is diameter, and what are its main features? Use a school-life example.
3. What is chords, and what are its main features? Use a real-world example.
4. What is circumference intuition, and what are its main features? Show the idea with a diagram where appropriate.
5. What is constructions., and what are its main features? Explain in your own words.
6. What is ■ Radius, and what are its main features? Give a step-by-step response.
7. What is diameter, and what are its main features? Mention one common misconception.
8. What is chords, and what are its main features? State one practical application.
9. What is circumference intuition, and what are its main features? Justify your response.
10. What is constructions., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Radius and explain each. Use a simple example.
12. Give two examples related to diameter and explain each. Use a school-life example.
13. Give two examples related to chords and explain each. Use a real-world example.
14. Give two examples related to circumference intuition and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to constructions. and explain each. Explain in your own words.
16. Give two examples related to ■ Radius and explain each. Give a step-by-step response.
17. Give two examples related to diameter and explain each. Mention one common misconception.
18. Give two examples related to chords and explain each. State one practical application.
19. Give two examples related to circumference intuition and explain each. Justify your response.
20. Give two examples related to constructions. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Radius related to diameter? Compare the two ideas. Use a simple example.
22. How is diameter related to chords? Compare the two ideas. Use a school-life example.
23. How is chords related to circumference intuition? Compare the two ideas. Use a real-world example.
24. How is circumference intuition related to constructions.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is constructions. related to ■ Radius? Compare the two ideas. Explain in your own words.
26. How is ■ Radius related to diameter? Compare the two ideas. Give a step-by-step response.
27. How is diameter related to chords? Compare the two ideas. Mention one common misconception.
28. How is chords related to circumference intuition? Compare the two ideas. State one practical application.
29. How is circumference intuition related to constructions.? Compare the two ideas. Justify your response.
30. How is constructions. related to ■ Radius? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Radius to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply diameter to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply chords to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply circumference intuition to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply constructions. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Radius to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply diameter to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply chords to a realistic situation. What would you do or calculate? State one practical application.
39. Apply circumference intuition to a realistic situation. What would you do or calculate? Justify your response.
40. Apply constructions. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Radius. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of diameter. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of chords. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of circumference intuition. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of constructions.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Radius. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of diameter. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of chords. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of circumference intuition. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of constructions.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Radius. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show diameter. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show chords. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show circumference intuition. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show constructions.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Radius. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show diameter. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show chords. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show circumference intuition. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show constructions.. Include the relevant rule or principle.

MATHEMATICS — 12. Ratio and Proportion

Coverage: ■ Ratios; equivalent ratios; unitary method; direct proportion; applications.

Understanding

1. What is ■ Ratios, and what are its main features? Use a simple example.
2. What is equivalent ratios, and what are its main features? Use a school-life example.
3. What is unitary method, and what are its main features? Use a real-world example.
4. What is direct proportion, and what are its main features? Show the idea with a diagram where appropriate.
5. What is applications., and what are its main features? Explain in your own words.
6. What is ■ Ratios, and what are its main features? Give a step-by-step response.
7. What is equivalent ratios, and what are its main features? Mention one common misconception.
8. What is unitary method, and what are its main features? State one practical application.
9. What is direct proportion, and what are its main features? Justify your response.
10. What is applications., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Ratios and explain each. Use a simple example.
12. Give two examples related to equivalent ratios and explain each. Use a school-life example.
13. Give two examples related to unitary method and explain each. Use a real-world example.
14. Give two examples related to direct proportion and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to applications. and explain each. Explain in your own words.
16. Give two examples related to ■ Ratios and explain each. Give a step-by-step response.
17. Give two examples related to equivalent ratios and explain each. Mention one common misconception.
18. Give two examples related to unitary method and explain each. State one practical application.
19. Give two examples related to direct proportion and explain each. Justify your response.
20. Give two examples related to applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Ratios related to equivalent ratios? Compare the two ideas. Use a simple example.
22. How is equivalent ratios related to unitary method? Compare the two ideas. Use a school-life example.
23. How is unitary method related to direct proportion? Compare the two ideas. Use a real-world example.
24. How is direct proportion related to applications.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is applications. related to ■ Ratios? Compare the two ideas. Explain in your own words.
26. How is ■ Ratios related to equivalent ratios? Compare the two ideas. Give a step-by-step response.
27. How is equivalent ratios related to unitary method? Compare the two ideas. Mention one common misconception.
28. How is unitary method related to direct proportion? Compare the two ideas. State one practical application.
29. How is direct proportion related to applications.? Compare the two ideas. Justify your response.
30. How is applications. related to ■ Ratios? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Ratios to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply equivalent ratios to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply unitary method to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply direct proportion to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply applications. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Ratios to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply equivalent ratios to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply unitary method to a realistic situation. What would you do or calculate? State one practical application.
39. Apply direct proportion to a realistic situation. What would you do or calculate? Justify your response.
40. Apply applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Ratios. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of equivalent ratios. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of unitary method. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of direct proportion. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Ratios. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of equivalent ratios. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of unitary method. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of direct proportion. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Ratios. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show equivalent ratios. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show unitary method. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show direct proportion. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Ratios. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show equivalent ratios. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show unitary method. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show direct proportion. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Include the relevant rule or principle.

MATHEMATICS — 13. Visualising Solid Shapes

Coverage: ■ Faces/edges/vertices; nets; views; 3D reasoning.

Understanding

1. What is ■ Faces/edges/vertices, and what are its main features? Use a simple example.
2. What is nets, and what are its main features? Use a school-life example.
3. What is views, and what are its main features? Use a real-world example.
4. What is 3D reasoning., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Faces/edges/vertices, and what are its main features? Explain in your own words.
6. What is nets, and what are its main features? Give a step-by-step response.
7. What is views, and what are its main features? Mention one common misconception.
8. What is 3D reasoning., and what are its main features? State one practical application.
9. What is ■ Faces/edges/vertices, and what are its main features? Justify your response.
10. What is nets, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to views and explain each. Use a simple example.
12. Give two examples related to 3D reasoning. and explain each. Use a school-life example.
13. Give two examples related to ■ Faces/edges/vertices and explain each. Use a real-world example.
14. Give two examples related to nets and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to views and explain each. Explain in your own words.
16. Give two examples related to 3D reasoning. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Faces/edges/vertices and explain each. Mention one common misconception.
18. Give two examples related to nets and explain each. State one practical application.
19. Give two examples related to views and explain each. Justify your response.
20. Give two examples related to 3D reasoning. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Faces/edges/vertices related to nets? Compare the two ideas. Use a simple example.
22. How is nets related to views? Compare the two ideas. Use a school-life example.
23. How is views related to 3D reasoning.? Compare the two ideas. Use a real-world example.
24. How is 3D reasoning. related to ■ Faces/edges/vertices? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Faces/edges/vertices related to nets? Compare the two ideas. Explain in your own words.
26. How is nets related to views? Compare the two ideas. Give a step-by-step response.
27. How is views related to 3D reasoning.? Compare the two ideas. Mention one common misconception.
28. How is 3D reasoning. related to ■ Faces/edges/vertices? Compare the two ideas. State one practical application.
29. How is ■ Faces/edges/vertices related to nets? Compare the two ideas. Justify your response.
30. How is nets related to views? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply views to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply 3D reasoning. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Faces/edges/vertices to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply nets to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply views to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply 3D reasoning. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Faces/edges/vertices to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply nets to a realistic situation. What would you do or calculate? State one practical application.
39. Apply views to a realistic situation. What would you do or calculate? Justify your response.
40. Apply 3D reasoning. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Faces/edges/vertices. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of nets. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of views. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of 3D reasoning.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Faces/edges/vertices. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of nets. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of views. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of 3D reasoning.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Faces/edges/vertices. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of nets. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show views. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show 3D reasoning.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Faces/edges/vertices. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show nets. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show views. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show 3D reasoning.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Faces/edges/vertices. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show nets. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show views. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show 3D reasoning.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 1. The Ever-Evolving World of Science

Coverage: ■ Nature of science; evidence; models; scientific progress and scientific temper.

Understanding

1. What is ■ Nature of science, and what are its main features? Use a simple example.
2. What is evidence, and what are its main features? Use a school-life example.
3. What is models, and what are its main features? Use a real-world example.
4. What is scientific progress and scientific temper., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Nature of science, and what are its main features? Explain in your own words.
6. What is evidence, and what are its main features? Give a step-by-step response.
7. What is models, and what are its main features? Mention one common misconception.
8. What is scientific progress and scientific temper., and what are its main features? State one practical application.
9. What is ■ Nature of science, and what are its main features? Justify your response.
10. What is evidence, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to models and explain each. Use a simple example.
12. Give two examples related to scientific progress and scientific temper. and explain each. Use a school-life example.
13. Give two examples related to ■ Nature of science and explain each. Use a real-world example.
14. Give two examples related to evidence and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to models and explain each. Explain in your own words.
16. Give two examples related to scientific progress and scientific temper. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Nature of science and explain each. Mention one common misconception.
18. Give two examples related to evidence and explain each. State one practical application.
19. Give two examples related to models and explain each. Justify your response.
20. Give two examples related to scientific progress and scientific temper. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Nature of science related to evidence? Compare the two ideas. Use a simple example.
22. How is evidence related to models? Compare the two ideas. Use a school-life example.
23. How is models related to scientific progress and scientific temper.? Compare the two ideas. Use a real-world example.
24. How is scientific progress and scientific temper. related to ■ Nature of science? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Nature of science related to evidence? Compare the two ideas. Explain in your own words.
26. How is evidence related to models? Compare the two ideas. Give a step-by-step response.
27. How is models related to scientific progress and scientific temper.? Compare the two ideas. Mention one common misconception.
28. How is scientific progress and scientific temper. related to ■ Nature of science? Compare the two ideas. State one practical application.
29. How is ■ Nature of science related to evidence? Compare the two ideas. Justify your response.
30. How is evidence related to models? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply models to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply scientific progress and scientific temper. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Nature of science to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply evidence to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply models to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply scientific progress and scientific temper. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Nature of science to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply evidence to a realistic situation. What would you do or calculate? State one practical application.
39. Apply models to a realistic situation. What would you do or calculate? Justify your response.
40. Apply scientific progress and scientific temper. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Nature of science. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of evidence. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of models. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of scientific progress and scientific temper.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Nature of science. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of evidence. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of models. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of scientific progress and scientific temper.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Nature of science. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of evidence. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show models. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show scientific progress and scientific temper.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Nature of science. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show evidence. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show models. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show scientific progress and scientific temper.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Nature of science. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show evidence. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show models. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show scientific progress and scientific temper.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 2. Exploring Substances: Acidic, Basic, and Neutral

Coverage: ■ Acids/bases; indicators; neutralisation; everyday substances.

Understanding

1. What is ■ Acids/bases, and what are its main features? Use a simple example.
2. What is indicators, and what are its main features? Use a school-life example.
3. What is neutralisation, and what are its main features? Use a real-world example.
4. What is everyday substances., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Acids/bases, and what are its main features? Explain in your own words.
6. What is indicators, and what are its main features? Give a step-by-step response.
7. What is neutralisation, and what are its main features? Mention one common misconception.
8. What is everyday substances., and what are its main features? State one practical application.
9. What is ■ Acids/bases, and what are its main features? Justify your response.
10. What is indicators, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to neutralisation and explain each. Use a simple example.
12. Give two examples related to everyday substances. and explain each. Use a school-life example.
13. Give two examples related to ■ Acids/bases and explain each. Use a real-world example.
14. Give two examples related to indicators and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to neutralisation and explain each. Explain in your own words.
16. Give two examples related to everyday substances. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Acids/bases and explain each. Mention one common misconception.
18. Give two examples related to indicators and explain each. State one practical application.
19. Give two examples related to neutralisation and explain each. Justify your response.
20. Give two examples related to everyday substances. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Acids/bases related to indicators? Compare the two ideas. Use a simple example.
22. How is indicators related to neutralisation? Compare the two ideas. Use a school-life example.
23. How is neutralisation related to everyday substances.? Compare the two ideas. Use a real-world example.
24. How is everyday substances. related to ■ Acids/bases? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Acids/bases related to indicators? Compare the two ideas. Explain in your own words.
26. How is indicators related to neutralisation? Compare the two ideas. Give a step-by-step response.
27. How is neutralisation related to everyday substances.? Compare the two ideas. Mention one common misconception.
28. How is everyday substances. related to ■ Acids/bases? Compare the two ideas. State one practical application.
29. How is ■ Acids/bases related to indicators? Compare the two ideas. Justify your response.
30. How is indicators related to neutralisation? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply neutralisation to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply everyday substances. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Acids/bases to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply indicators to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply neutralisation to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply everyday substances. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Acids/bases to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply indicators to a realistic situation. What would you do or calculate? State one practical application.
39. Apply neutralisation to a realistic situation. What would you do or calculate? Justify your response.
40. Apply everyday substances. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Acids/bases. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of indicators. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of neutralisation. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of everyday substances.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Acids/bases. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of indicators. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of neutralisation. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of everyday substances.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Acids/bases. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of indicators. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show neutralisation. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show everyday substances.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Acids/bases. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show indicators. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show neutralisation. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show everyday substances.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Acids/bases. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show indicators. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show neutralisation. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show everyday substances.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 3. Electricity: Circuits and their Components

Coverage: ■ Cells; circuits; conductors/insulators; symbols; series ideas.

Understanding

1. What is ■ Cells, and what are its main features? Use a simple example.
2. What is circuits, and what are its main features? Use a school-life example.
3. What is conductors/insulators, and what are its main features? Use a real-world example.
4. What is symbols, and what are its main features? Show the idea with a diagram where appropriate.
5. What is series ideas., and what are its main features? Explain in your own words.
6. What is ■ Cells, and what are its main features? Give a step-by-step response.
7. What is circuits, and what are its main features? Mention one common misconception.
8. What is conductors/insulators, and what are its main features? State one practical application.
9. What is symbols, and what are its main features? Justify your response.
10. What is series ideas., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Cells and explain each. Use a simple example.
12. Give two examples related to circuits and explain each. Use a school-life example.
13. Give two examples related to conductors/insulators and explain each. Use a real-world example.
14. Give two examples related to symbols and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to series ideas. and explain each. Explain in your own words.
16. Give two examples related to ■ Cells and explain each. Give a step-by-step response.
17. Give two examples related to circuits and explain each. Mention one common misconception.
18. Give two examples related to conductors/insulators and explain each. State one practical application.
19. Give two examples related to symbols and explain each. Justify your response.
20. Give two examples related to series ideas. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Cells related to circuits? Compare the two ideas. Use a simple example.
22. How is circuits related to conductors/insulators? Compare the two ideas. Use a school-life example.
23. How is conductors/insulators related to symbols? Compare the two ideas. Use a real-world example.
24. How is symbols related to series ideas.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is series ideas. related to ■ Cells? Compare the two ideas. Explain in your own words.
26. How is ■ Cells related to circuits? Compare the two ideas. Give a step-by-step response.
27. How is circuits related to conductors/insulators? Compare the two ideas. Mention one common misconception.
28. How is conductors/insulators related to symbols? Compare the two ideas. State one practical application.
29. How is symbols related to series ideas.? Compare the two ideas. Justify your response.
30. How is series ideas. related to ■ Cells? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Cells to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply circuits to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply conductors/insulators to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply symbols to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply series ideas. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Cells to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply circuits to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply conductors/insulators to a realistic situation. What would you do or calculate? State one practical application.
39. Apply symbols to a realistic situation. What would you do or calculate? Justify your response.
40. Apply series ideas. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Cells. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of circuits. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of conductors/insulators. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of symbols. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of series ideas.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Cells. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of circuits. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of conductors/insulators. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of symbols. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of series ideas.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Cells. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show circuits. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show conductors/insulators. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show symbols. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show series ideas.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Cells. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show circuits. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show conductors/insulators. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show symbols. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show series ideas.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 4. The World of Metals and Non-metals

Coverage: ■ Physical/chemical properties; uses; reactivity patterns; materials.

Understanding

1. What is ■ Physical/chemical properties, and what are its main features? Use a simple example.
2. What is uses, and what are its main features? Use a school-life example.
3. What is reactivity patterns, and what are its main features? Use a real-world example.
4. What is materials., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Physical/chemical properties, and what are its main features? Explain in your own words.
6. What is uses, and what are its main features? Give a step-by-step response.
7. What is reactivity patterns, and what are its main features? Mention one common misconception.
8. What is materials., and what are its main features? State one practical application.
9. What is ■ Physical/chemical properties, and what are its main features? Justify your response.
10. What is uses, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to reactivity patterns and explain each. Use a simple example.
12. Give two examples related to materials. and explain each. Use a school-life example.
13. Give two examples related to ■ Physical/chemical properties and explain each. Use a real-world example.
14. Give two examples related to uses and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to reactivity patterns and explain each. Explain in your own words.
16. Give two examples related to materials. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Physical/chemical properties and explain each. Mention one common misconception.
18. Give two examples related to uses and explain each. State one practical application.
19. Give two examples related to reactivity patterns and explain each. Justify your response.
20. Give two examples related to materials. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Physical/chemical properties related to uses? Compare the two ideas. Use a simple example.
22. How is uses related to reactivity patterns? Compare the two ideas. Use a school-life example.
23. How is reactivity patterns related to materials.? Compare the two ideas. Use a real-world example.
24. How is materials. related to ■ Physical/chemical properties? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Physical/chemical properties related to uses? Compare the two ideas. Explain in your own words.
26. How is uses related to reactivity patterns? Compare the two ideas. Give a step-by-step response.
27. How is reactivity patterns related to materials.? Compare the two ideas. Mention one common misconception.
28. How is materials. related to ■ Physical/chemical properties? Compare the two ideas. State one practical application.
29. How is ■ Physical/chemical properties related to uses? Compare the two ideas. Justify your response.
30. How is uses related to reactivity patterns? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply reactivity patterns to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply materials. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Physical/chemical properties to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply uses to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply reactivity patterns to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply materials. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Physical/chemical properties to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply uses to a realistic situation. What would you do or calculate? State one practical application.
39. Apply reactivity patterns to a realistic situation. What would you do or calculate? Justify your response.
40. Apply materials. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Physical/chemical properties. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of uses. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of reactivity patterns. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of materials.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Physical/chemical properties. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of uses. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of reactivity patterns. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of materials.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Physical/chemical properties. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of uses. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show reactivity patterns. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show materials.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Physical/chemical properties. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show uses. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show reactivity patterns. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show materials.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Physical/chemical properties. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show uses. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show reactivity patterns. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show materials.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 5. Changes Around Us: Physical and Chemical

Coverage: NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus

Understanding

1. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a simple example.
2. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a school-life example.
3. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a real-world example.
4. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Show the idea with a diagram where appropriate.
5. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Explain in your own words.
6. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Give a step-by-step response.
7. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Mention one common misconception.
8. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? State one practical application.
9. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Justify your response.
10. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a simple example.
12. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a school-life example.
13. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a real-world example.
14. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Explain in your own words.
16. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Give a step-by-step response.
17. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Mention one common misconception.
18. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. State one practical application.
19. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Justify your response.
20. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Include the relevant rule or principle.

Comparison

21. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a simple example.
22. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a school-life example.

- 23.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a real-world example.
- 24.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Show the idea with a diagram where appropriate.
- 25.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Explain in your own words.
- 26.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Give a step-by-step response.
- 27.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Mention one common misconception.
- 28.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. State one practical application.
- 29.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Justify your response.
- 30.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Include the relevant rule or principle.

Application

- 31.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a simple example.
- 32.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a school-life example.
- 33.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a real-world example.
- 34.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
- 35.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Explain in your own words.
- 36.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Give a step-by-step response.
- 37.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Mention one common misconception.
- 38.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? State one practical application.
- 39.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Justify your response.
- 40.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

- 41.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a simple example.
- 42.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a school-life example.
- 43.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a real-world example.
- 44.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
- 45.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Explain in your own words.
- 46.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Give a step-by-step response.
- 47.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 6. Adolescence: A Stage of Growth and Change

Coverage: ■ Puberty; physical/emotional changes; hormones; health and hygiene.

Understanding

1. What is ■ Puberty, and what are its main features? Use a simple example.
2. What is physical/emotional changes, and what are its main features? Use a school-life example.
3. What is hormones, and what are its main features? Use a real-world example.
4. What is health and hygiene., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Puberty, and what are its main features? Explain in your own words.
6. What is physical/emotional changes, and what are its main features? Give a step-by-step response.
7. What is hormones, and what are its main features? Mention one common misconception.
8. What is health and hygiene., and what are its main features? State one practical application.
9. What is ■ Puberty, and what are its main features? Justify your response.
10. What is physical/emotional changes, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to hormones and explain each. Use a simple example.
12. Give two examples related to health and hygiene. and explain each. Use a school-life example.
13. Give two examples related to ■ Puberty and explain each. Use a real-world example.
14. Give two examples related to physical/emotional changes and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to hormones and explain each. Explain in your own words.
16. Give two examples related to health and hygiene. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Puberty and explain each. Mention one common misconception.
18. Give two examples related to physical/emotional changes and explain each. State one practical application.
19. Give two examples related to hormones and explain each. Justify your response.
20. Give two examples related to health and hygiene. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Puberty related to physical/emotional changes? Compare the two ideas. Use a simple example.
22. How is physical/emotional changes related to hormones? Compare the two ideas. Use a school-life example.
23. How is hormones related to health and hygiene.? Compare the two ideas. Use a real-world example.
24. How is health and hygiene. related to ■ Puberty? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Puberty related to physical/emotional changes? Compare the two ideas. Explain in your own words.
26. How is physical/emotional changes related to hormones? Compare the two ideas. Give a step-by-step response.
27. How is hormones related to health and hygiene.? Compare the two ideas. Mention one common misconception.
28. How is health and hygiene. related to ■ Puberty? Compare the two ideas. State one practical application.
29. How is ■ Puberty related to physical/emotional changes? Compare the two ideas. Justify your response.
30. How is physical/emotional changes related to hormones? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply hormones to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply health and hygiene. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Puberty to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply physical/emotional changes to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply hormones to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply health and hygiene. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Puberty to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply physical/emotional changes to a realistic situation. What would you do or calculate? State one practical application.
39. Apply hormones to a realistic situation. What would you do or calculate? Justify your response.
40. Apply health and hygiene. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Puberty. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of physical/emotional changes. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of hormones. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of health and hygiene.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Puberty. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of physical/emotional changes. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of hormones. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of health and hygiene.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Puberty. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of physical/emotional changes. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show hormones. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show health and hygiene.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Puberty. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show physical/emotional changes. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show hormones. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show health and hygiene.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Puberty. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show physical/emotional changes. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show hormones. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show health and hygiene.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 7. Heat Transfer in Nature

Coverage: ■ Conduction; convection; radiation; everyday applications.

Understanding

1. What is ■ Conduction, and what are its main features? Use a simple example.
2. What is convection, and what are its main features? Use a school-life example.
3. What is radiation, and what are its main features? Use a real-world example.
4. What is everyday applications., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Conduction, and what are its main features? Explain in your own words.
6. What is convection, and what are its main features? Give a step-by-step response.
7. What is radiation, and what are its main features? Mention one common misconception.
8. What is everyday applications., and what are its main features? State one practical application.
9. What is ■ Conduction, and what are its main features? Justify your response.
10. What is convection, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to radiation and explain each. Use a simple example.
12. Give two examples related to everyday applications. and explain each. Use a school-life example.
13. Give two examples related to ■ Conduction and explain each. Use a real-world example.
14. Give two examples related to convection and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to radiation and explain each. Explain in your own words.
16. Give two examples related to everyday applications. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Conduction and explain each. Mention one common misconception.
18. Give two examples related to convection and explain each. State one practical application.
19. Give two examples related to radiation and explain each. Justify your response.
20. Give two examples related to everyday applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Conduction related to convection? Compare the two ideas. Use a simple example.
22. How is convection related to radiation? Compare the two ideas. Use a school-life example.
23. How is radiation related to everyday applications.? Compare the two ideas. Use a real-world example.
24. How is everyday applications. related to ■ Conduction? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Conduction related to convection? Compare the two ideas. Explain in your own words.
26. How is convection related to radiation? Compare the two ideas. Give a step-by-step response.
27. How is radiation related to everyday applications.? Compare the two ideas. Mention one common misconception.
28. How is everyday applications. related to ■ Conduction? Compare the two ideas. State one practical application.
29. How is ■ Conduction related to convection? Compare the two ideas. Justify your response.
30. How is convection related to radiation? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply radiation to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply everyday applications. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Conduction to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply convection to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply radiation to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply everyday applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Conduction to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply convection to a realistic situation. What would you do or calculate? State one practical application.
39. Apply radiation to a realistic situation. What would you do or calculate? Justify your response.
40. Apply everyday applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Conduction. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of convection. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of radiation. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of everyday applications.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Conduction. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of convection. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of radiation. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of everyday applications.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Conduction. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of convection. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show radiation. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show everyday applications.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Conduction. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show convection. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show radiation. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show everyday applications.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Conduction. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show convection. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show radiation. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show everyday applications.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 8. Measurement of Time and Motion

Coverage: ■ Time; distance; speed; motion representation and measurement.

Understanding

1. What is ■ Time, and what are its main features? Use a simple example.
2. What is distance, and what are its main features? Use a school-life example.
3. What is speed, and what are its main features? Use a real-world example.
4. What is motion representation and measurement., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Time, and what are its main features? Explain in your own words.
6. What is distance, and what are its main features? Give a step-by-step response.
7. What is speed, and what are its main features? Mention one common misconception.
8. What is motion representation and measurement., and what are its main features? State one practical application.
9. What is ■ Time, and what are its main features? Justify your response.
10. What is distance, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to speed and explain each. Use a simple example.
12. Give two examples related to motion representation and measurement. and explain each. Use a school-life example.
13. Give two examples related to ■ Time and explain each. Use a real-world example.
14. Give two examples related to distance and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to speed and explain each. Explain in your own words.
16. Give two examples related to motion representation and measurement. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Time and explain each. Mention one common misconception.
18. Give two examples related to distance and explain each. State one practical application.
19. Give two examples related to speed and explain each. Justify your response.
20. Give two examples related to motion representation and measurement. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Time related to distance? Compare the two ideas. Use a simple example.
22. How is distance related to speed? Compare the two ideas. Use a school-life example.
23. How is speed related to motion representation and measurement.? Compare the two ideas. Use a real-world example.
24. How is motion representation and measurement. related to ■ Time? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Time related to distance? Compare the two ideas. Explain in your own words.
26. How is distance related to speed? Compare the two ideas. Give a step-by-step response.
27. How is speed related to motion representation and measurement.? Compare the two ideas. Mention one common misconception.
28. How is motion representation and measurement. related to ■ Time? Compare the two ideas. State one practical application.
29. How is ■ Time related to distance? Compare the two ideas. Justify your response.
30. How is distance related to speed? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply speed to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply motion representation and measurement. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Time to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply distance to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply speed to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply motion representation and measurement. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Time to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply distance to a realistic situation. What would you do or calculate? State one practical application.
39. Apply speed to a realistic situation. What would you do or calculate? Justify your response.
40. Apply motion representation and measurement. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Time. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of speed. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of motion representation and measurement.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Time. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of speed. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of motion representation and measurement.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Time. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show speed. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show motion representation and measurement.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Time. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show distance. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show speed. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show motion representation and measurement.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Time. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show distance. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show speed. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show motion representation and measurement.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 9. Life Processes in Animals

Coverage: ■ Nutrition; digestion; respiration; circulation; excretion.

Understanding

1. What is ■ Nutrition, and what are its main features? Use a simple example.
2. What is digestion, and what are its main features? Use a school-life example.
3. What is respiration, and what are its main features? Use a real-world example.
4. What is circulation, and what are its main features? Show the idea with a diagram where appropriate.
5. What is excretion., and what are its main features? Explain in your own words.
6. What is ■ Nutrition, and what are its main features? Give a step-by-step response.
7. What is digestion, and what are its main features? Mention one common misconception.
8. What is respiration, and what are its main features? State one practical application.
9. What is circulation, and what are its main features? Justify your response.
10. What is excretion., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Nutrition and explain each. Use a simple example.
12. Give two examples related to digestion and explain each. Use a school-life example.
13. Give two examples related to respiration and explain each. Use a real-world example.
14. Give two examples related to circulation and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to excretion. and explain each. Explain in your own words.
16. Give two examples related to ■ Nutrition and explain each. Give a step-by-step response.
17. Give two examples related to digestion and explain each. Mention one common misconception.
18. Give two examples related to respiration and explain each. State one practical application.
19. Give two examples related to circulation and explain each. Justify your response.
20. Give two examples related to excretion. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Nutrition related to digestion? Compare the two ideas. Use a simple example.
22. How is digestion related to respiration? Compare the two ideas. Use a school-life example.
23. How is respiration related to circulation? Compare the two ideas. Use a real-world example.
24. How is circulation related to excretion.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is excretion. related to ■ Nutrition? Compare the two ideas. Explain in your own words.
26. How is ■ Nutrition related to digestion? Compare the two ideas. Give a step-by-step response.
27. How is digestion related to respiration? Compare the two ideas. Mention one common misconception.
28. How is respiration related to circulation? Compare the two ideas. State one practical application.
29. How is circulation related to excretion.? Compare the two ideas. Justify your response.
30. How is excretion. related to ■ Nutrition? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Nutrition to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply digestion to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply respiration to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply circulation to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply excretion. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Nutrition to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply digestion to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply respiration to a realistic situation. What would you do or calculate? State one practical application.
39. Apply circulation to a realistic situation. What would you do or calculate? Justify your response.
40. Apply excretion. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Nutrition. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of digestion. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of respiration. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of circulation. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of excretion.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Nutrition. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of digestion. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of respiration. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of circulation. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of excretion.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Nutrition. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show digestion. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show respiration. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show circulation. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show excretion.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Nutrition. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show digestion. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show respiration. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show circulation. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show excretion.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 10. Life Processes in Plants

Coverage: ■ Photosynthesis; respiration; transport; transpiration.

Understanding

1. What is ■ Photosynthesis, and what are its main features? Use a simple example.
2. What is respiration, and what are its main features? Use a school-life example.
3. What is transport, and what are its main features? Use a real-world example.
4. What is transpiration., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Photosynthesis, and what are its main features? Explain in your own words.
6. What is respiration, and what are its main features? Give a step-by-step response.
7. What is transport, and what are its main features? Mention one common misconception.
8. What is transpiration., and what are its main features? State one practical application.
9. What is ■ Photosynthesis, and what are its main features? Justify your response.
10. What is respiration, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to transport and explain each. Use a simple example.
12. Give two examples related to transpiration. and explain each. Use a school-life example.
13. Give two examples related to ■ Photosynthesis and explain each. Use a real-world example.
14. Give two examples related to respiration and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to transport and explain each. Explain in your own words.
16. Give two examples related to transpiration. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Photosynthesis and explain each. Mention one common misconception.
18. Give two examples related to respiration and explain each. State one practical application.
19. Give two examples related to transport and explain each. Justify your response.
20. Give two examples related to transpiration. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Photosynthesis related to respiration? Compare the two ideas. Use a simple example.
22. How is respiration related to transport? Compare the two ideas. Use a school-life example.
23. How is transport related to transpiration.? Compare the two ideas. Use a real-world example.
24. How is transpiration. related to ■ Photosynthesis? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Photosynthesis related to respiration? Compare the two ideas. Explain in your own words.
26. How is respiration related to transport? Compare the two ideas. Give a step-by-step response.
27. How is transport related to transpiration.? Compare the two ideas. Mention one common misconception.
28. How is transpiration. related to ■ Photosynthesis? Compare the two ideas. State one practical application.
29. How is ■ Photosynthesis related to respiration? Compare the two ideas. Justify your response.
30. How is respiration related to transport? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply transport to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply transpiration. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Photosynthesis to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply respiration to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply transport to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply transpiration. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Photosynthesis to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply respiration to a realistic situation. What would you do or calculate? State one practical application.
39. Apply transport to a realistic situation. What would you do or calculate? Justify your response.
40. Apply transpiration. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Photosynthesis. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of respiration. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of transport. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of transpiration.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Photosynthesis. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of respiration. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of transport. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of transpiration.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Photosynthesis. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of respiration. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show transport. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show transpiration.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Photosynthesis. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show respiration. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show transport. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show transpiration.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Photosynthesis. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show respiration. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show transport. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show transpiration.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 11. Light: Shadows and Reflections

Coverage: ■ Light propagation; shadows; reflection; mirrors; ray-based reasoning.

Understanding

1. What is ■ Light propagation, and what are its main features? Use a simple example.
2. What is shadows, and what are its main features? Use a school-life example.
3. What is reflection, and what are its main features? Use a real-world example.
4. What is mirrors, and what are its main features? Show the idea with a diagram where appropriate.
5. What is ray-based reasoning., and what are its main features? Explain in your own words.
6. What is ■ Light propagation, and what are its main features? Give a step-by-step response.
7. What is shadows, and what are its main features? Mention one common misconception.
8. What is reflection, and what are its main features? State one practical application.
9. What is mirrors, and what are its main features? Justify your response.
10. What is ray-based reasoning., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Light propagation and explain each. Use a simple example.
12. Give two examples related to shadows and explain each. Use a school-life example.
13. Give two examples related to reflection and explain each. Use a real-world example.
14. Give two examples related to mirrors and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to ray-based reasoning. and explain each. Explain in your own words.
16. Give two examples related to ■ Light propagation and explain each. Give a step-by-step response.
17. Give two examples related to shadows and explain each. Mention one common misconception.
18. Give two examples related to reflection and explain each. State one practical application.
19. Give two examples related to mirrors and explain each. Justify your response.
20. Give two examples related to ray-based reasoning. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Light propagation related to shadows? Compare the two ideas. Use a simple example.
22. How is shadows related to reflection? Compare the two ideas. Use a school-life example.
23. How is reflection related to mirrors? Compare the two ideas. Use a real-world example.
24. How is mirrors related to ray-based reasoning.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ray-based reasoning. related to ■ Light propagation? Compare the two ideas. Explain in your own words.
26. How is ■ Light propagation related to shadows? Compare the two ideas. Give a step-by-step response.
27. How is shadows related to reflection? Compare the two ideas. Mention one common misconception.
28. How is reflection related to mirrors? Compare the two ideas. State one practical application.
29. How is mirrors related to ray-based reasoning.? Compare the two ideas. Justify your response.
30. How is ray-based reasoning. related to ■ Light propagation? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Light propagation to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply shadows to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply reflection to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply mirrors to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply ray-based reasoning. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Light propagation to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply shadows to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply reflection to a realistic situation. What would you do or calculate? State one practical application.
39. Apply mirrors to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ray-based reasoning. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Light propagation. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of shadows. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of reflection. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of mirrors. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ray-based reasoning.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Light propagation. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of shadows. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of reflection. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of mirrors. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of ray-based reasoning.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Light propagation. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show shadows. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show reflection. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show mirrors. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ray-based reasoning.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Light propagation. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show shadows. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show reflection. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show mirrors. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show ray-based reasoning.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 12. Earth, Moon, and the Sun

Coverage: ■ Earth-Moon-Sun system; phases; eclipses; day/night; seasons at an introductory level.

Understanding

1. What is ■ Earth-Moon-Sun system, and what are its main features? Use a simple example.
2. What is phases, and what are its main features? Use a school-life example.
3. What is eclipses, and what are its main features? Use a real-world example.
4. What is day/night, and what are its main features? Show the idea with a diagram where appropriate.
5. What is seasons at an introductory level., and what are its main features? Explain in your own words.
6. What is ■ Earth-Moon-Sun system, and what are its main features? Give a step-by-step response.
7. What is phases, and what are its main features? Mention one common misconception.
8. What is eclipses, and what are its main features? State one practical application.
9. What is day/night, and what are its main features? Justify your response.
10. What is seasons at an introductory level., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Earth-Moon-Sun system and explain each. Use a simple example.
12. Give two examples related to phases and explain each. Use a school-life example.
13. Give two examples related to eclipses and explain each. Use a real-world example.
14. Give two examples related to day/night and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to seasons at an introductory level. and explain each. Explain in your own words.
16. Give two examples related to ■ Earth-Moon-Sun system and explain each. Give a step-by-step response.
17. Give two examples related to phases and explain each. Mention one common misconception.
18. Give two examples related to eclipses and explain each. State one practical application.
19. Give two examples related to day/night and explain each. Justify your response.
20. Give two examples related to seasons at an introductory level. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Earth-Moon-Sun system related to phases? Compare the two ideas. Use a simple example.
22. How is phases related to eclipses? Compare the two ideas. Use a school-life example.
23. How is eclipses related to day/night? Compare the two ideas. Use a real-world example.
24. How is day/night related to seasons at an introductory level.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is seasons at an introductory level. related to ■ Earth-Moon-Sun system? Compare the two ideas. Explain in your own words.
26. How is ■ Earth-Moon-Sun system related to phases? Compare the two ideas. Give a step-by-step response.
27. How is phases related to eclipses? Compare the two ideas. Mention one common misconception.
28. How is eclipses related to day/night? Compare the two ideas. State one practical application.
29. How is day/night related to seasons at an introductory level.? Compare the two ideas. Justify your response.
30. How is seasons at an introductory level. related to ■ Earth-Moon-Sun system? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Earth-Moon-Sun system to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply phases to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply eclipses to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply day/night to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply seasons at an introductory level. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Earth-Moon-Sun system to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply phases to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply eclipses to a realistic situation. What would you do or calculate? State one practical application.
39. Apply day/night to a realistic situation. What would you do or calculate? Justify your response.
40. Apply seasons at an introductory level. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Earth-Moon-Sun system. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of phases. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of eclipses. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of day/night. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of seasons at an introductory level.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Earth-Moon-Sun system. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of phases. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of eclipses. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of day/night. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of seasons at an introductory level.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Earth-Moon-Sun system. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show phases. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show eclipses. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show day/night. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show seasons at an introductory level.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Earth-Moon-Sun system. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show phases. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show eclipses. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show day/night. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show seasons at an introductory level.. Include the relevant rule or principle.